

Dicke-ladder factorization of finite-time quantum Otto-engine power

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Abstract

We identify a Dicke-ladder factorization of finite-time power in a collectively dissipative quantum Otto engine. In the high-temperature bulk, the contact protocol and the efficiency at maximum power are selected by a single-particle finite-time kernel, while the output scale is fixed by the collective Dicke-ladder variance. Consequently, the known stationary Dicke multiplier, $(n + 2)/3$, becomes a finite-time multiplier for limit-cycle power, maximum power, and the inverse relative variance of the running output. For fixed contact-time ratios the first correction in contact duration is quadratic, not linear. The short-contact efficiency at maximum power has a closed form independent of n , while the optimized power retains the same Dicke gain. We also distinguish the finite-temperature ceiling of the high-temperature expansion from the cold-edge reset scale $n_* \sim 1/(\gamma\tau)$, showing that the latter is not a bulk cutoff of the Dicke factor.

I. INTRODUCTION

Collective and many-body quantum heat engines have been studied from several angles [1–3, 6–15, 24]. Latune, Sinayskiy, and Petruccione [1] showed that collective bath coupling changes the steady-state heat capacity of a spin ensemble. Near the high-temperature Carnot point, for spin-1/2 media, the stationary work gain is $G_W^{\text{stat}} \rightarrow (n + 2)/3$, from the Dicke-ladder variance $K_n = n(n + 2)/12$ versus $K_1 = 1/4$ for a single spin [30]. We take that stationary factor as known.

Finite-time operation could spoil this simplicity. The engine state is then the *limit cycle*, not a prepared thermal state, the hot and cold maps are applied for finite times, and power must be optimized over the protocol. Kloc, Cejnar, and Schaller [3] analyzed collective Otto cycles with coherent spin dissipation, including the approach to the limit cycle and superradiant relaxation. Our question is sharper: in the high-temperature bulk, does the stationary Dicke variance remain an operational finite-time resource, or is it replaced by a protocol-dependent collective effect? Related collective reliability under shared baths is in Ref. [9]; efficiency at maximum power for few-body Otto engines is an old subject [16–23, 25, 26]. Fast two-level Otto cycles were treated by Erdman *et al.* [4] and more generally by Cavina *et al.* [5]; at $n = 1$ the leading finite-time correction need not be linear.

We find that, for the Dicke ladder in the high-temperature bulk, the variance mechanism survives in a stronger form. The finite-time problem factorizes: the contact protocol, the

maximum-power frequency ratio, and the running-output fluctuation kernel are selected by single-particle finite-time dynamics, while the output scale is fixed by K_n . Thus $(n+2)/3$ multiplies limit-cycle power at short and long contact, maximum power, and the inverse relative work variance. For fixed contact ratios the first correction in contact duration is quadratic. The scale $n\gamma\tau$ is not a bulk cutoff of this factorization; it enters only when the optimized cold stroke reaches a cold-edge reset regime. Finite temperature instead limits the high- T expansion through $n\theta \sim 1$.

II. MODEL AND FIGURE OF MERIT

Consider n two-level atoms restricted to the fully symmetric Dicke sector $j = n/2$ [30], with $H(\omega) = \omega J_z$. Collective contact with a thermal bath of occupation $\bar{n} = [e^{\omega/T} - 1]^{-1}$ produces a birth-death process on the Dicke populations. In the excitation coordinate $r = 0, \dots, n$ the rates are $\lambda_r = \gamma\bar{n}(n-r)(r+1)$ and $\mu_r = \gamma(1+\bar{n})r(n-r+1)$. On this sector, and for populations alone, the dynamics is classical. The quantum content of the model is the Dicke matrix element that fixes those rates and the variance K_n . The enhancement we discuss is a collective-sector effect; we do not use coherences. The question below is whether this variance remains only a stationary heat-capacity mechanism, or whether it also acts as the multiplicative resource for finite-time limit-cycle power, maximum power, and output stability.

We take a sudden Otto cycle [17, 25, 26]. The frequency is quenched between ω_c and ω_h , and the system is held for times τ_h and τ_c in contact with the hot and cold baths, with generators $\mathcal{W}_h = \mathcal{W}(\omega_h, T_h)$ and $\mathcal{W}_c = \mathcal{W}(\omega_c, T_c)$. The work strokes commute with J_z , so only populations enter the energy balance. The efficiency is then fixed at $\eta = 1 - \omega_c/\omega_h$.

One cycle of the populations is the map $M = e^{\mathcal{W}_c\tau_c}e^{\mathcal{W}_h\tau_h}$. We use the limit cycle (periodic steady state) $Mp_{\text{LC}} = p_{\text{LC}}$ as the engine state. The collective power gain is $G_P = P_n/(nP_1)$, where P_1 is the power of a single two-level engine with the same protocol and the same bare γ .

We call this the running-output reliability because it is the inverse relative variance of the accumulated work in the stationary limit cycle, $\mathcal{R}_{\text{run}} = \langle W \rangle_{\text{LC}}^2 / V_{\text{run}}$, with V_{run} the long-time variance per cycle (appendices). For n independent two-level engines the benchmark is $n\mathcal{R}_{\text{run}}^{(1)}$. It should not be confused with a finite prepared-state window, which contains

startup modes of M and is used only as a diagnostic [31]. The Supplemental Material also gives the single-cycle version, where one samples one cycle from the limit-cycle ensemble. The analytical results refer to the clean collective ladder. If the symmetry is broken, the invariant sector changes. A few small- n checks of that kind are given in the Supplemental Material [29, 31].

III. FINITE-TIME LIMIT-CYCLE FACTORIZATION

We first show that the same variance mechanism controls the two clean limit-cycle endpoints: the mixed-generator short-contact limit and the full-thermalization long-contact limit. By the high-temperature bulk we mean $\bar{n}_h, \bar{n}_c \gg n$, so that the thermal weight sits well inside the ladder rather than at an edge. For equal contacts $\tau_h = \tau_c = \tau$ the short-time map is $e^{\mathcal{W}_c \tau} e^{\mathcal{W}_h \tau} = \exp[\tau(\mathcal{W}_h + \mathcal{W}_c) + O(\tau^2)]$, and the limit cycle tends to the steady state of the mixed generator $\mathcal{L}_0 = \mathcal{W}_h + \mathcal{W}_c$. That mixed generator is again a detailed-balance ladder, with $e^{-\theta_{\text{eff}}} = (\bar{n}_h + \bar{n}_c)/(2 + \bar{n}_h + \bar{n}_c)$. At leading high temperature the short-contact power is $P_{\text{LC}}(0) = \gamma(\omega_h - \omega_c)K_n(\bar{n}_h - \bar{n}_c)/(\bar{n}_h + \bar{n}_c)$, and therefore

$$G_P^{\text{LC}}(\tau \rightarrow 0) = \frac{K_n}{nK_1} = \frac{n+2}{3} + O(n^2\theta^2), \quad (1)$$

while the long-contact limit is full thermalization on each isochore, $G_P^{\text{LC}}(\tau \rightarrow \infty) = G_W^{\text{stat}}$. Thus the finite-time kernel is fixed by single-particle rates and bath occupations, whereas the full n -dependence enters through K_n . The same leading factor appears at both ends. The correction $O(n^2\theta^2)$ is a finite-temperature effect in the moment expansion; it becomes important when $n\theta \sim 1$, and should not be confused with the cold-edge scale $n\gamma\tau$ treated later. Even at finite θ the short-time power is known in closed form: it is the hot moment current in the mixed Gibbs state at θ_{eff} . As a check, for $\omega_h = 10$, $t = 10^{-3}$, and $n = 256$ this expression gives $G_P^{\text{LC}} = 72.814$, against the numerical limit-cycle value 72.81. In deep high temperature the short-contact EMP factorization $G_P^* = (n+2)/3$ remains accurate through $n = 500$ (relative error $\sim 10^{-4}$; Supplemental Material [31]).

We also find that there is no linear correction in the contact time when the two contacts scale together. Write $\tau_h = \alpha\tau$, $\tau_c = \beta\tau$ with fixed $\alpha, \beta > 0$ and $\alpha + \beta = 1$. Then $M(\tau) = e^{B\beta\tau} e^{A\alpha\tau}$ with $A = \mathcal{W}_h$, $B = \mathcal{W}_c$, and $L_0 = \alpha A + \beta B$. Expanding the fixed point

of M and matching powers of τ (appendices; Supplemental Material [31]), one obtains

$$P_{\text{LC}}(\tau) = P_{\text{LC}}(0) + O(\tau^2), \quad (2)$$

or $G_P^{\text{LC}}(\tau) = G_P^{(0)}[1 - a_n\gamma\tau + O((\gamma\tau)^2)]$ with $a_n = 0$ and $G_P^{(0)} = (n+2)/3 + O(n^2\theta^2)$. For simplicity the figures and the appendices use equal contacts; the fixed-ratio case is treated in the Supplemental Material. If the ratio τ_h/τ_c itself depends on τ , a linear term is not ruled out by this expansion. This cancellation is a property of the fixed-ratio product structure of the cycle map. The Dicke structure enters only in the value of $P_{\text{LC}}(0)$, not in the absence of the linear term. In the bulk, a smooth change of G_P over a wide range of τ is then the interpolation between the mixed and stationary endpoints, not a linear small- τ collapse [Fig. 1].

IV. FACTORIZED MAXIMUM-POWER OPTIMIZATION

We keep the bath temperatures fixed, write $\eta_C = 1 - T_c/T_h$ and $t = T_c/T_h = 1 - \eta_C$. In the high-temperature short-contact limit, maximizing over the frequency ratio $x = \omega_c/\omega_h \in (t, 1)$ reduces to a single-particle problem because the power factorizes as $P = \gamma K_n \omega_h f(x)$ with $f(x) = (1-x)(x-t)/(x+t)$. The maximizer is unique, $x^* = -t + \sqrt{2t(1+t)}$, and the efficiency at maximum power is

$$\eta_{\text{short}}^* = 1 + t - \sqrt{2t(1+t)}, \quad (3)$$

or $\eta_{\text{short}}^* = 2 - \eta_C - \sqrt{2(1-\eta_C)(2-\eta_C)}$. The peak power is $P_{\text{max}}^{\text{short}} = \gamma K_n \omega_h [1 + 3t - 2\sqrt{2t(1+t)}]$. Since K_n sits outside $f(x)$, the optimum x^* does not depend on n , and

$$G_P^* = \frac{P_{\text{max}}^{(n)}}{nP_{\text{max}}^{(1)}} = \frac{n+2}{3}. \quad (4)$$

The collective content is the factorization, not only the closed form of η_{short}^* . The function $f(x)$ is the same single-particle finite-time kernel for every n , whereas the height of the power curve carries K_n . Thus the optimizer x^* is selected without reference to n , while the optimized power is collectively amplified by the Dicke-ladder variance. This is the finite-time mechanism behind $G_P^* = (n+2)/3$. The same holds if the contact ratio is optimized as well at fixed total cycle time (Supplemental Material [31]). Figure 3(a) shows $P/(\gamma K_n \omega_h)$ for several n falling on $f(x)$. Panel (b) shows that the normalized gain at the optimum stays near one in the bulk and falls when $n\theta_h \sim 1$.

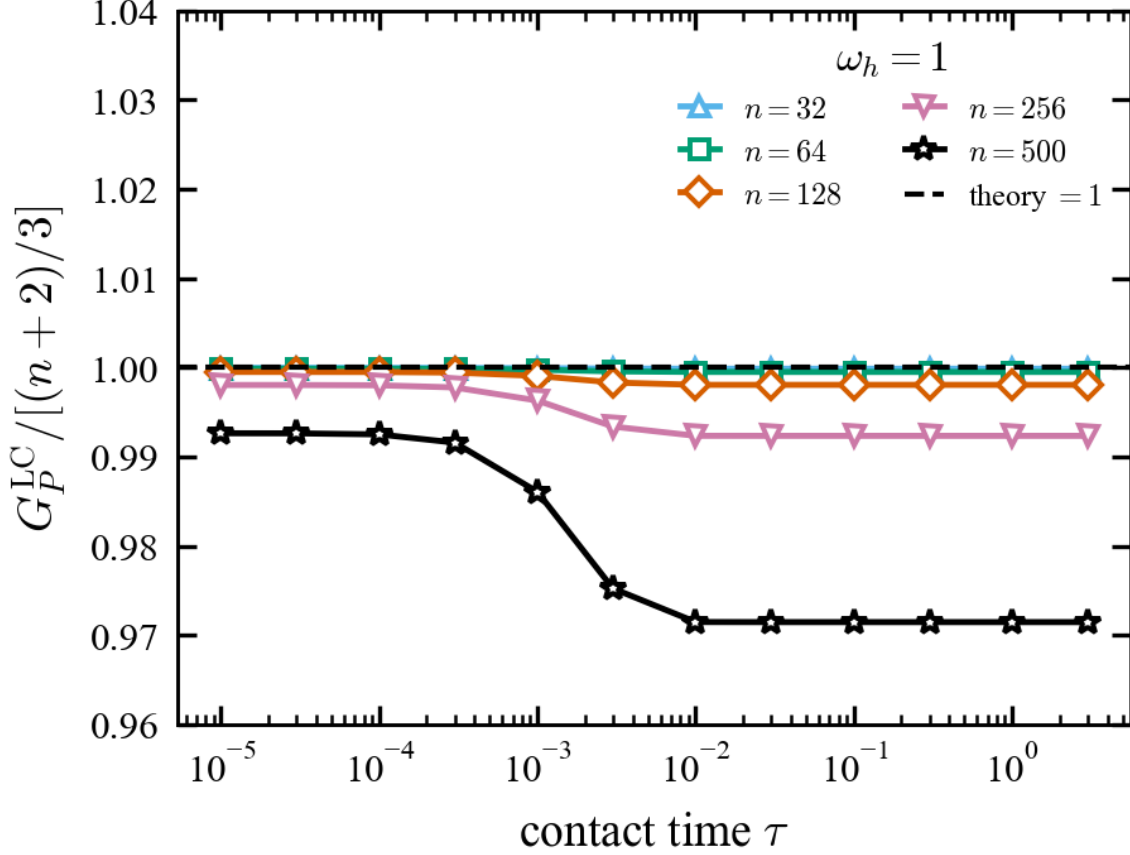


FIG. 1. **Limit-cycle factorization in the high-temperature bulk.** Normalized limit-cycle gain $G_P^{\text{LC}}/[(n+2)/3]$ against contact time at $\omega_h = 1$, $t = 10^{-3}$, for $n = 32, 64, 128, 256, 500$. At short contact the curves sit on the Dicke value through $n = 500$; a mild long- τ depression grows with n and is a finite-temperature/edge residual, not a bulk $n\gamma\tau$ collapse of the short-time factor. Finite-window averages from a prepared state can underestimate the gain because they contain startup modes of M ; see the Supplemental Material [31] and the appendices.

If both contacts are long enough for full thermalization, maximizing the stationary Otto work at high temperature recovers the standard result $\eta_{\text{long}}^* = \eta_C/2$ [16, 17], again with $G_P^* = (n+2)/3$. Thus the two analytic maximum-power limits have the same structure: different kernels select different optimal frequency ratios, but the collective scale of the optimized power is the same Dicke-ladder variance. In both limits the power has the form $P = \Theta\eta(\eta_C - \eta)$ with $\Theta \propto K_n$. Collectivity multiplies the coefficient of the power–efficiency curve but does not take the engine to Carnot at finite power [27, 28]. For small t , $\eta_{\text{short}}^* = 1 - \sqrt{2t} + O(t)$ lies below Curzon–Ahlborn, $1 - \sqrt{t}$ [18, 27].

V. FINITE-TIME RELIABILITY

The same factorization controls output stability. For a stochastic cycle $W = \Delta\omega(m_C - m_A)$ sampled on the limit cycle, the Supplemental Material shows that, in the high-temperature bulk, both the mean work and the accumulated-work variance per cycle factorize into K_n times finite-time functions that are independent of n . Consequently the inverse relative variance of the running output obeys

$$\frac{\mathcal{R}_{\text{run}}^{(n)}}{n\mathcal{R}_{\text{run}}^{(1)}} = \frac{K_n}{nK_1} = \frac{n+2}{3} \quad (5)$$

within the same high-temperature bulk regime. Absolute $\mathcal{R}_{\text{run}}$ still depends on the contact times and vanishes as $\tau \rightarrow 0$, but the collective *gain* remains fixed by K_n . Exact tilted-map calculations of the accumulated-work cumulants confirm $\mathcal{R}_{\text{run}}^{(n)}/[n\mathcal{R}_{\text{run}}^{(1)}] = (n+2)/3$ through $n = 128$ at short contact, long contact, and the short-contact EMP point [Fig. 2; Supplemental Material [31]]. The single-cycle diagnostic $\mathcal{R}_{\text{cyc}} = \langle W \rangle^2 / \text{Var}(W)$ obeys the same leading gain. For the periodically sampled work current, the same high-temperature reduction gives the TUR-type product $\Sigma/\mathcal{R}_{\text{run}} = 2$. The same factor K_n that enhances the mean output also appears in the entropy production, so the collective stability gain is compatible with the corresponding uncertainty trade-off rather than a parametric evasion of it.

VI. EMP PATH AND COLD-EDGE BOUNDARY

The factorized endpoints are connected by the intermediate-contact maximum-power path. In deep high temperature the optimal frequency ratio moves from the short-contact anchor x^* to the long-contact anchor $x^{**} = (1+t)/2$, while $G_P^*(\tau)$ stays close to $(n+2)/3$ along the path [Fig. 4(a),(b)]. The location of the optimum is again essentially single-particle; the output scale is Dicke.

The scale $n\gamma\tau$ is familiar from collective relaxation and superradiance. The question is where it controls the engine. In the high-temperature bulk the limit-cycle gain is set by K_n . A different regime appears when the optimized cold stroke leaves that bulk ($\bar{n}_c \ll n$). Reset is then controlled by $s = n\gamma\tau$, and the limit-cycle power collapses as

$$P \propto \gamma n(n+2) \frac{F_{\text{eff}}(s)}{s}, \quad (6)$$

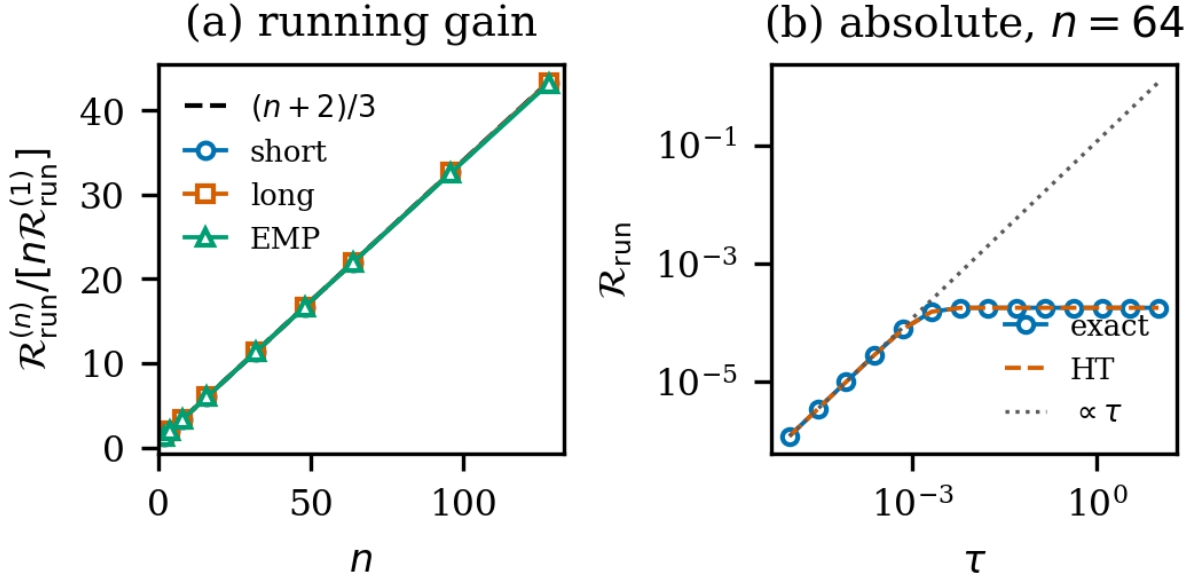


FIG. 2. **Factorized running-output reliability.** (a) Exact running-output inverse-relative-variance gain $\mathcal{R}_{\text{run}}^{(n)}/[n\mathcal{R}_{\text{run}}^{(1)}]$ on the limit cycle, computed from the dominant eigenvalue of the tilted one-cycle map, at short contact, long contact, and the short-contact EMP point; dashed line, $(n+2)/3$. (b) Absolute running reliability versus contact time at fixed $n = 64$, showing that $\mathcal{R}_{\text{run}} \propto \tau$ at short contact while the normalized collective gain remains Dicke. The corresponding single-cycle diagnostic $\mathcal{R}_{\text{cyc}}$ is given in the Supplemental Material.

where F_{eff} is the distribution-averaged cold-reset scaling function derived in the Supplemental Material. This is a large- n mean-field description, supported by a collapse of the finite- n data onto $F_{\text{eff}}(s)/s$. We do not claim an exact finite- n theorem. The local size exponent crosses from roughly two to one around $s \sim 1$. This identifies the crossover scale

$$n_* \sim \frac{1}{\gamma\tau}. \quad (7)$$

The scale $n\theta \sim 1$ is separate: it marks the failure of the high- T moment expansion, not a finite-time cutoff. Thus $n\gamma\tau$ is not an intrinsic cutoff of the high-temperature bulk factorization; it becomes the controlling variable only after the optimized cold stroke leaves the bulk and enters a cold-edge reset regime [Fig. 4(c),(d)].

VII. CONCLUSIONS

We have identified a Dicke-ladder factorization of finite-time quantum Otto-engine power in the high-temperature bulk. The known stationary Dicke factor $(n+2)/3$ multiplies the limit-cycle power at both contact endpoints. For fixed contact ratios the first correction in contact time is quadratic. At short contact, maximum-power optimization gives $\eta_{\text{short}}^* = 1 + t - \sqrt{2t(1+t)}$, with $t = T_c/T_h = 1 - \eta_C$; the optimizer is independent of n , and $G_P^* = (n+2)/3$. At long contact one recovers $\eta_C/2$ with the same gain. Finite temperature limits the useful high-temperature expansion through $n\theta \sim 1$. The scale $n_* \sim 1/(\gamma\tau)$ appears only when the cold stroke enters the cold-edge reset regime. The same high-temperature reduction gives $\mathcal{R}_{\text{run}}^{(n)}/[n\mathcal{R}_{\text{run}}^{(1)}] = (n+2)/3$, confirmed by exact tilted-map accumulated-work cumulants through $n = 128$. Thus the Dicke factor survives finite-time operation as both a power multiplier and a stability multiplier. These statements refer to the collectively coupled symmetric Dicke ladder: representative checks suggest that loss of collectivity, rather than local dephasing within the symmetric-sector dynamics, is the relevant fragility.

Appendix A: Running-output stability

The running reliability used in the main text is the inverse relative variance of the accumulated work in the limit cycle, $\mathcal{R}_{\text{run}} = \langle W \rangle_{\text{LC}}^2 / V_{\text{run}}$, with

$$V_{\text{run}} = \lim_{N \rightarrow \infty} \frac{1}{N} \text{Var} \left(\sum_{k=0}^{N-1} W_k \right).$$

In the high-temperature bulk the mean coordinate relaxes under bath $b = h, c$ as

$$\begin{aligned} \langle m \rangle &\mapsto a_b \langle m \rangle + (1 - a_b) m_b^{\text{eq}}, \\ a_b &= e^{-\Gamma_b \tau_b}, \quad m_b^{\text{eq}} = -K_n \theta_b. \end{aligned}$$

Solving the two-stroke limit cycle gives

$$\langle W \rangle_{\text{LC}} = \Delta\omega K_n (\theta_c - \theta_h) \frac{(1 - a_h)(1 - a_c)}{1 - a_h a_c}.$$

The stationary covariance sum gives, to the same order,

$$V_{\text{run}}^{(n)} = 2(\Delta\omega)^2 K_n \frac{(1 - a_h)(1 - a_c)}{1 - a_h a_c}.$$

Here $\Gamma_b \simeq 2\gamma\bar{n}_b$ in the high-temperature bulk. Thus $\langle W \rangle_{\text{LC}}$ and V_{run} each carry one factor of the collective variance K_n . After forming $\mathcal{R}_{\text{run}} = \langle W \rangle_{\text{LC}}^2 / V_{\text{run}}$, one net factor of K_n remains, giving the normalized gain $K_n / (nK_1) = (n+2)/3$. The single-cycle diagnostic is $\mathcal{R}_{\text{cyc}} = \langle W \rangle^2 / \text{Var}(W)$ for one cycle sampled from the limit cycle. The Supplemental Material [31] gives the tilted-map construction used for exact $\mathcal{R}_{\text{run}}$, retains $\mathcal{R}_{\text{cyc}}$ as a diagnostic, and summarizes the numerical checks in Fig. 2.

Appendix B: Cancellation of the linear finite-time correction

We treat the equal-contact case in full. The fixed-ratio argument ($\tau_h = \alpha\tau$, $\tau_c = \beta\tau$) is given in the Supplemental Material [31]. Let $A = \mathcal{W}_h$, $B = \mathcal{W}_c$, $L_0 = A + B$, and $M = I + \tau L_0 + \tau^2 C + O(\tau^3)$ with $C = \frac{1}{2}(A^2 + B^2 + 2BA) = \frac{1}{2}L_0^2 + \frac{1}{2}[B, A]$. Write the limit cycle as $p_\tau = p_0 + \tau p_1 + \tau^2 p_2 + O(\tau^3)$, with $\mathbf{1}^\top p_0 = 1$ and $\mathbf{1}^\top p_k = 0$ for $k \geq 1$. Then $Mp_\tau = p_\tau$ becomes

$$\begin{aligned} \mathbf{1} : \quad & p_0 + \tau(p_1 + L_0 p_0) + \tau^2(p_2 + L_0 p_1 + C p_0) + O(\tau^3) \\ & = p_0 + \tau p_1 + \tau^2 p_2 + O(\tau^3). \end{aligned} \tag{B1}$$

At order τ one finds only $L_0 p_0 = 0$: the same τp_1 appears on both sides and cancels, so p_1 is not forced to vanish. At order τ^2 , $L_0 p_1 + C p_0 = 0$. Using $L_0 p_0 = 0$ this is $L_0 p_1 = -\frac{1}{2}[B, A]p_0$ (p_2 drops out and is fixed only at higher order). From $Bp_0 = -Ap_0$ one has $[B, A]p_0 = L_0 Ap_0$, so $L_0(p_1 + \frac{1}{2}Ap_0) = 0$. For an irreducible detailed-balance ladder, $\ker L_0 = \text{span}\{p_0\}$ on the simplex, and $\mathbf{1}^\top Ap_0 = 0$ gives

$$p_1 = -\frac{1}{2}Ap_0 \quad (p_1 \neq 0 \text{ in general}). \tag{B2}$$

Here Ap_0 is the hot flow on the mixed steady state; it vanishes only if the two baths share the same stationary state.

The cycle work is $W_{\text{LC}}(\tau) = (\omega_h - \omega_c)u^\top(e^{A\tau} - I)p_\tau$ with $u_m = m$. Expanding $e^{A\tau} - I$ yields

$$\begin{aligned} W_{\text{LC}}(\tau) = (\omega_h - \omega_c) & \left[\tau u^\top Ap_0 \right. \\ & \left. + \tau^2 \left(u^\top Ap_1 + \frac{1}{2}u^\top A^2 p_0 \right) + O(\tau^3) \right]. \end{aligned} \tag{B3}$$

With Eq. (B2) the coefficient of τ^2 cancels, so $P_{\text{LC}} = W_{\text{LC}}/(2\tau)$ satisfies

$$P_{\text{LC}}(\tau) = \frac{\omega_h - \omega_c}{2} u^\top A p_0 + O(\tau^2), \quad (\text{B4})$$

which is Eq. (2). Setting $p_1 = 0$ by hand, or trying to cancel the linear power with p_2 , does not match the fixed-point equations (Supplemental Material [31]).

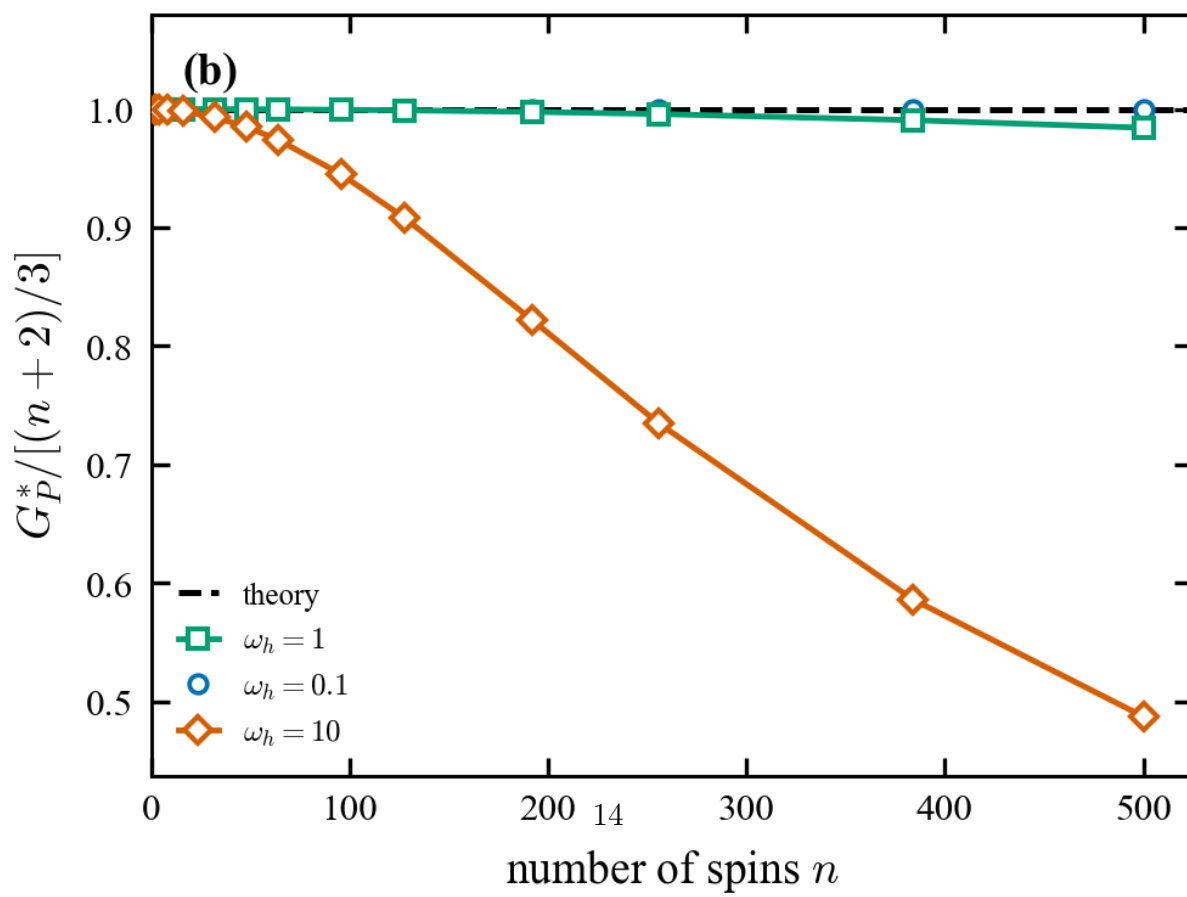
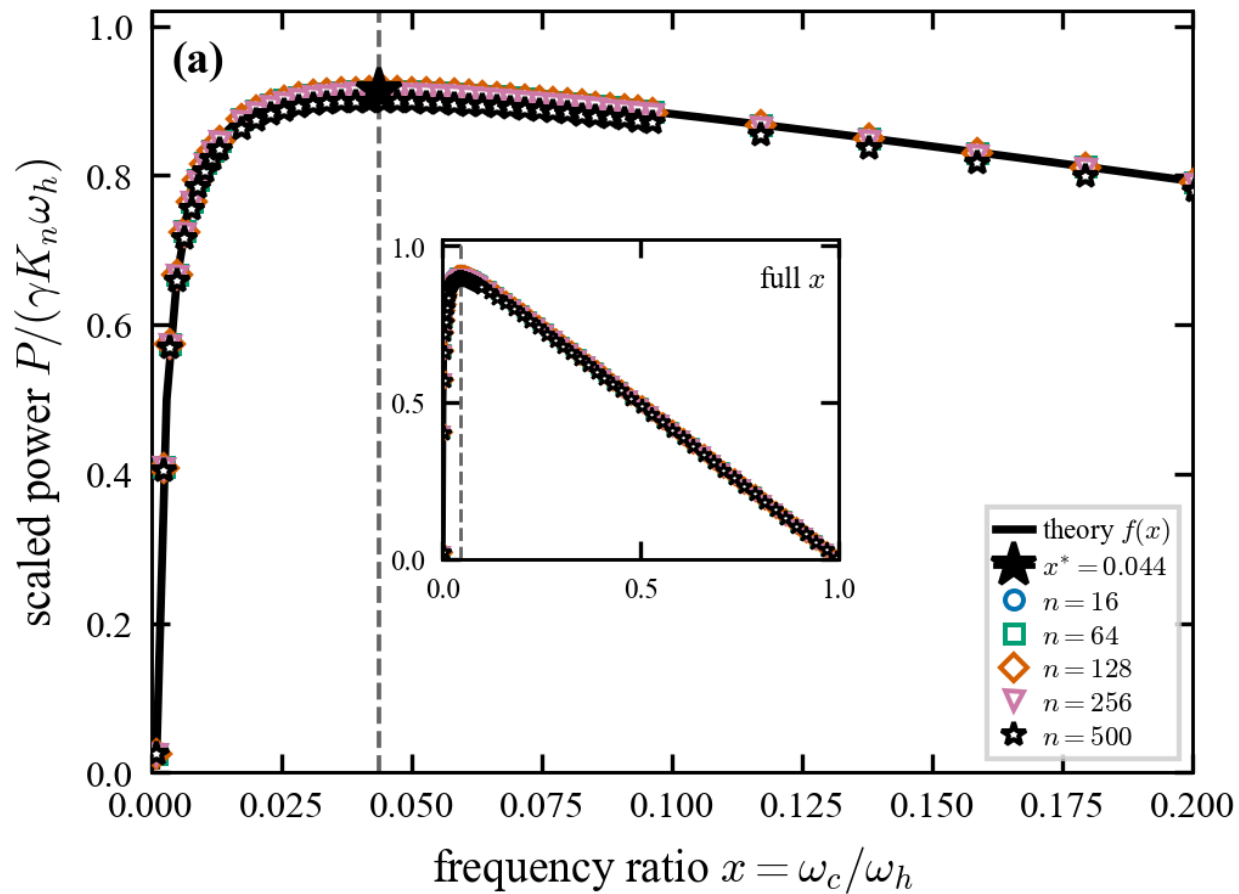
Appendix C: Startup windows

If one runs only N cycles from an initial state p_{in} and M is diagonalizable, with one-cycle work linear in the initial population, then $W_N = W_{\text{LC}} + N^{-1} \sum_{r \geq 2} a_r (1 - \lambda_r^N)/(1 - \lambda_r)$, where λ_r are the nonstationary eigenvalues of M . Such a window therefore mixes startup with engine operation. Near Carnot the single-spin stationary work is very small, and a ground-state first cycle can dominate the average and pull G_P down.

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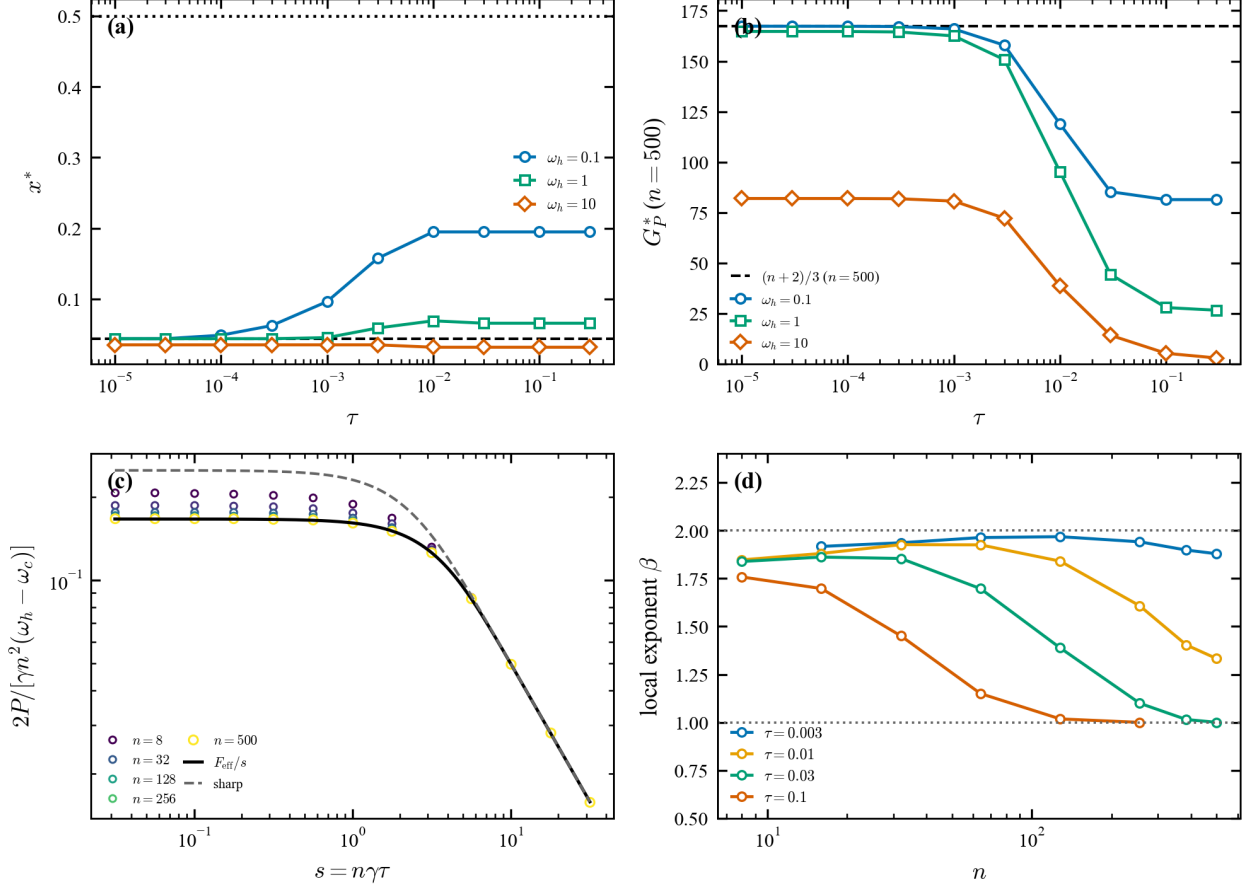


FIG. 4. **Factorized EMP path and cold-edge boundary.** (a) In deep high temperature the optimized frequency ratio migrates from the short-contact anchor to the long-contact anchor ($n = 500$ limit cycle). (b) Along this migration the EMP gain remains near the Dicke value $(n+2)/3$ in the bulk; at larger frequency scale the gain drops only after the optimized cold stroke leaves the high-temperature bulk. (c) In the cold-edge regime the scaled limit-cycle power for $n = 8$ – 500 collapses as a function of $s = n\gamma\tau$, matching $F_{\text{eff}}(s)/s$. (d) The local exponent crosses from $\beta \simeq 2$ to $\beta \simeq 1$, consistent with the crossover scale $n_* \sim 1/(\gamma\tau)$.